

TRAINING DAY

CBPCA's hands-on training opens the eyes of would-be home performance contractors.

BY ELKA KARL

Contractors in Northern and Central California who are interested in learning about home performance contracting have a great resource in their backyard. The California Building Performance Contractors Association (CBPCA) is a non-profit organization that is charged with introducing remodeling contractors—and the general public—to home performance services. Funding comes from California energy ratepayers under the auspices of the California Public Utilities Commission. Since it started training activities in late 2002, CBPCA has trained more than 100 individuals from nearly 65 different firms.

As part of its ongoing series of trainings, the CBPCA recently held a field class at Carol Markell's home in the Kensington neighborhood, near Berkeley. Markell, who is the advertising and marketing manager at *Home Energy*, had run into Steve Sokolsky from CBPCA at a building performance conference. When she found out that her home could be used as a test house for one of CBPCA's trainings, she jumped at the chance to get her home evaluated.

The CBPCA's overall training regimen consists of three parts: business and marketing; diagnosis and remediation; and one-on-one field mentoring. The one-day business and marketing session provides an overview of home performance contracting, commonsense marketing strategies, and examples of how the selling of home performance services is different from traditional sales.

The six-day diagnosis and remediation session is the meat of the program. The first three days are spent in the classroom covering building science, combustion safety, and diagnostic tools and methods. The next two days are



CBPCA trainer Rick Chitwood checks blower door test results with his class.

spent in the field, performing diagnoses on actual customer homes such as Carol Markell's. The final day is spent back in the classroom, developing workscopes for the homes diagnosed in the field.

A few weeks after the initial meeting, trainers Timothy Locke and Rick Chitwood showed up with about a dozen class members to run Markell's house through its paces. After performing diagnostic tests, including infrared imaging and blower door tests, they told Markell that her house needed help. Markell knew that her house wasn't as tight as it could be, but it wasn't until after the class that she knew what she was dealing with. "It confirmed my suspicions that my home was leaky, but I didn't know how bad," says Markell. "Air circulating

from underneath, leaks in the canned lighting and all the outlets, a hole in the floor that I never saw." Based on the results of the training class, Markell plans to have her home sealed and a new heater installed by next fall—both verbal recommendations from trainers Locke and Chitwood.

Whole House Learning

Although Markell definitely learned a lot about her house on that day, the real beneficiaries of whole-house knowledge were the building professionals who took the class. One of the class participants, Karam Khalsa, the owner-contractor of Springwood Builders in Lafayette, California, has

been in the building profession for more than 25 years. He works primarily with residential remodeling, building additions, whole-house remodels, kitchens, and baths. Although he has done green building jobs, he has only recently begun work in home performance contracting.

His interest was piqued through the Affordable Comfort Conference in San Ramon—the same conference, incidentally, where Markell convinced Sokolsky to use her house as a test home. “I had received a few flyers and e-mails,” says Khalsa, “and the whole-house systems approach seemed intriguing.”

Integrating building performance into his business seemed like an obvious next step for Khalsa. “Treating the home as a whole system—an ecosystem you might say—fits right in with green building, which I have been very interested and active in for quite some time. Besides conserving energy, I am particularly interested in the home environment, indoor air quality, and the very important need to create homes that are healthier and more comfortable for us to live in and where we like to spend our time, while at the same time allowing us to contribute to the support of our fragile planet. The building performance program is adding a very valuable element to my business as a remodeling contractor.”

Khalsa found the training to be well worth his time, and was impressed by the extensive and detailed information covered in the program. “The trainers, Tim and Rick, were excellent. They presented the program in a well-organized way, so that the ideas were easy to follow.” The class participants first learned about the principles and concepts of the whole-house system. After that, they stepped into the test homes to use the diagnostic tools they’d been learning about. “Seeing the results firsthand, then and there, was a dramatic learning experience,” says Khalsa.

Khalsa plans on using the information he learned at the CBPCA training at Springwood Builders, and will integrate home performance diagnostics and remediation into his business. He will also incorporate construction procedures and design features based on

whole-house performance principles when diagnostics aren’t performed. “The home performance approach and diagnostics will help me and my team in upcoming remodeling projects,” he says.

Another participant in the program, Pam Molsick, who was previously employed with environmental and energy consulting firms, now works for Sustainable Spaces, a company that provides building performance improvement services for residential and commercial customers.

Like Khalsa, Molsick found the training very helpful. “I actually perform many of the diagnostic tests that they taught us. I use the concepts they taught us every day to figure out how a home got so moldy, cold, or expensive to heat. It helped me understand how all of the systems work together.” Molsick found the hands-on approach to the class very effective—and surprising. “We were all amazed at how much energy was being lost through the floor, walls, ceiling, and ducts. When you look through the infrared camera and see it, you believe it. Tim and Rick have been contractors for a very long time. Their on-the-job personal experiences really were eye-opening. I also learned from others taking the class. Many contractors were astounded that they had been building homes the wrong way for 20 years.”

Molsick is such a firm believer in building science that she’s made it her entire business. She is LEED accredited, and believes that the holistic approach to examining buildings—using the whole-house approach—is the best way to make sure all of the systems are working as a functioning unit. “What you do to one part of the home affects another part of the home,” she says. “For example, if an electrician puts wires over an insulation batt instead of in between the insulation batts, the R-value of the insulation batt can be seriously compromised and create convection currents, which can make a home uncomfortable. If a plumber does not seal the plumbing properly or a contractor misses a fire-stop, this can also create leakage in the home. Electricians, plumbers, HVAC specialists, and contractors seem to focus only on their part of the home and do not con-

sider energy efficiency while they are building. After they take this class, they will realize they affect more than just the electrical or plumbing work, but the whole home.”

So would these two participants in the CBPCA class recommend the training to other contractors? The answer is a definite yes from both. “We—contractors, homeowners, and our communities—will all benefit from treating homes as whole systems and incorporating new building science ideas,” says Khalsa.

After completing the diagnosis and remediation training, Khalsa, Molsick, and the other students have the opportunity to receive one-on-one field mentoring from Locke and Chitwood. The trainers will accompany them on their initial diagnoses and help them work out any technical, business, or marketing problems they may encounter. This individual attention is especially valuable to contractors who are new to diagnostics, or to those wishing to change their company into a building performance firm.

Molsick predicts that it will be both financially and environmentally beneficial for contractors to incorporate building science techniques into their businesses. “Things are rapidly changing in home development,” she says. “Soon cities and states will want to see building performance testing from contractors when they turn in their permit applications. It would behoove people to get the training or hire building performance consultants if they want to be leaders in the market.”



*Elka Karl is an associate editor at **Home Energy**.*

FOR MORE INFORMATION:

To learn more about the CBPCA's training program, go to www.cbPCA.org.